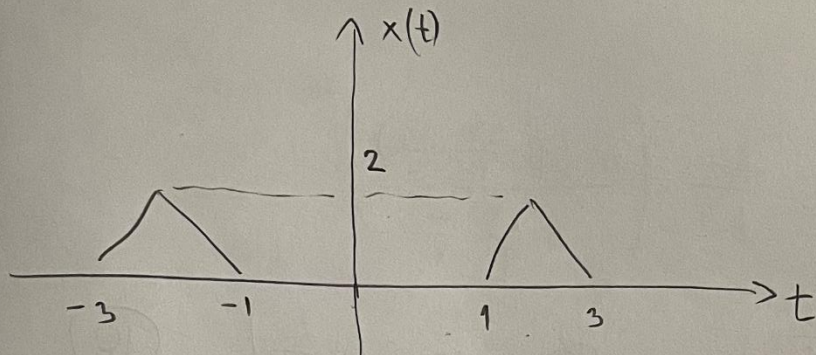


1.

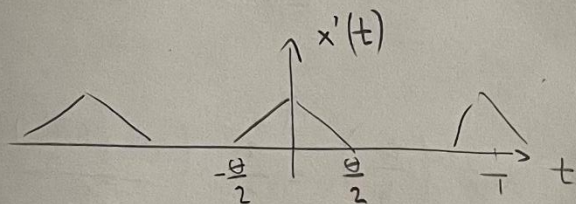


$$T = 4 \text{ ms}$$

$$\theta = 2 \text{ ms}$$

$$L = \frac{\theta}{T} = \frac{1}{2}$$

$$N = \frac{2}{L} = 4$$



$$x'(t) = \frac{a_0}{2} + \sum_{n=1}^{+\infty} a_n \cdot \cos(n\omega_0 t) + \sum_{n=1}^{+\infty} b_n \cdot \sin(n\omega_0 t) \quad (2)$$

$$b_n = 0 \quad (1)$$

$$a_n = \frac{2}{T} \int_{-\frac{\theta}{2}}^{\frac{\theta}{2}} x'(t) \cdot \cos(n\omega_0 t) dt = \frac{2}{T} \int_{-\frac{\theta}{2}}^{\frac{\theta}{2}} x'(t) \cdot \cos(n\omega_0 t) dt = \frac{4}{T} \int_0^{\frac{\theta}{2}} x'(t) \cos(n\omega_0 t) dt$$

$$1^o \quad (0, A) \quad \Rightarrow \quad x' = -\frac{2A}{\theta} \left(t - \frac{\theta}{2} \right) = -\frac{2A}{\theta} t + A$$

$$2^o \quad \left(\frac{\theta}{2}, 0 \right)$$

$$a_n = \frac{4}{T} \int_0^{\frac{\theta}{2}} \left(-\frac{2A}{\theta} t + A \right) \cdot \cos(n\omega_0 t) dt = -\frac{8A}{\theta T} \int_0^{\frac{\theta}{2}} t \cdot \cos(n\omega_0 t) dt + \frac{4A}{T} \int_0^{\frac{\theta}{2}} \cos(n\omega_0 t) dt =$$

$$\left. \begin{array}{l} u=t \quad dv=\cos(n\omega_0 t) dt \\ du=dt \quad v=\frac{1}{n\omega_0} \cdot \sin(n\omega_0 t) \end{array} \right|_0^{\frac{\theta}{2}}$$

$$= -\frac{8A}{\theta T} \cdot \left[\frac{t}{n\omega_0} \cdot \sin(n\omega_0 t) + \frac{1}{(n\omega_0)^2} \cdot \cos(n\omega_0 t) \right] \Big|_0^{\frac{\theta}{2}} + \frac{4A}{n\omega_0 T} \cdot \sin(n\omega_0 t) \Big|_0^{\frac{\theta}{2}} =$$

$$= -\frac{8A}{\theta T} \left[\frac{\theta}{2n\omega_0} \cdot \sin(n\omega_0 \cdot \frac{\theta}{2}) + \frac{1}{(n\omega_0)^2} \left(\cos(n\omega_0 \cdot \frac{\theta}{2}) - 1 \right) \right] + \frac{4A}{n\omega_0 T} \cdot \sin(n\omega_0 \cdot \frac{\theta}{2}) =$$

$$= -\frac{4A}{n\omega_0 T} \cdot \sin(n\omega_0 \cdot \frac{\theta}{2}) + \frac{8A}{\theta T \cdot (n\omega_0)^2} \left(1 - \cos(n\omega_0 \cdot \frac{\theta}{2}) \right) + \frac{4A}{n\omega_0 T} \cdot \sin(n\omega_0 \cdot \frac{\theta}{2}) =$$

$$= \frac{8A}{\theta T \cdot (n\omega_0)^2} \cdot 2 \cdot \sin^2(n\omega_0 \cdot \frac{\theta}{4}) = \frac{16A}{\theta T} \cdot \frac{\sin^2(n\omega_0 \cdot \frac{\theta}{4})}{(n\omega_0)^2} \cdot \frac{\theta}{16} \cdot \frac{16}{\theta} =$$

$$= \frac{A\theta}{T} \cdot \left(\frac{\sin(n\omega_0 \frac{\theta}{4})}{n\omega_0 \frac{\theta}{4}} \right)^2$$

$$\underline{F_u}' = \frac{a_n - j b_n}{2} = \frac{a_n}{2} = \frac{A\theta}{2T} \cdot \left(\frac{\sin(n \cdot \frac{2\pi}{T_0} \cdot \frac{\theta}{4})}{n \cdot \frac{2\pi}{T_0} \cdot \frac{\theta}{4}} \right)^2 =$$

$$= \frac{A\omega_0}{2} \cdot \left(\frac{\sin(n\omega_0 \frac{L}{2})}{n\omega_0 \frac{L}{2}} \right)^2 = \frac{A\omega_0}{2} \cdot \text{sinc}^2\left(n \frac{L}{2}\right) \quad (10)$$

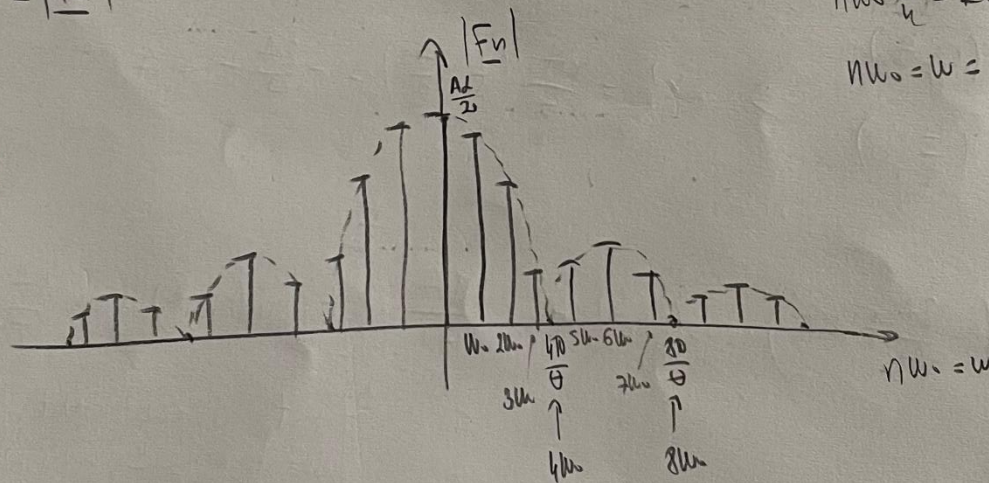
$$\left. \begin{array}{l} L = \frac{1}{2} \\ A = 2 \end{array} \right\} \Rightarrow \underline{F_u}' = \frac{1}{2} \cdot \text{sinc}^2\left(\frac{n}{4}\right) = \frac{1}{2} \cdot \left(\frac{\sin(n\omega_0/4)}{n\omega_0/4} \right)^2$$

$$\underline{F_u} = \underline{F_u}' \cdot e^{-jn\omega_0 t_0} = \frac{A\omega_0}{2} \cdot \text{sinc}^2\left(n \frac{L}{2}\right) \cdot e^{-jn\omega_0 t_0} \quad (5)$$

$$t_0 = 2\text{ms}$$

$$|\underline{F_u}'| = |\underline{F_u}|$$

(5)

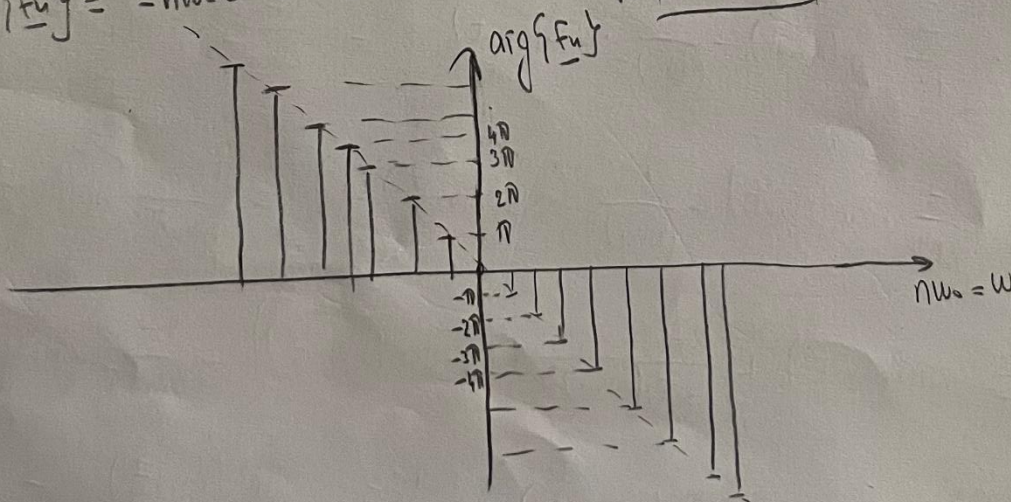


$$n\omega_0 \frac{\theta}{4} = k\pi$$

$$\begin{aligned} n\omega_0 = \omega &= \frac{4\pi}{\theta} \cdot k = \frac{4\pi}{T} \cdot \frac{T}{\theta} \cdot k = \\ &= 2\omega_0 \cdot \frac{1}{2} \cdot k = \\ &= 4\omega_0 \cdot k \end{aligned}$$

$$\arg\{\underline{F_u}\} = -n\omega_0 t_0 = -2n\omega_0 = -2n \cdot \frac{2\pi}{T} = -n\pi$$

(5)



$$F_0 = \frac{A L}{2} = \frac{1}{2}$$

$$F_1 = \frac{A L}{2} \cdot \frac{\sin^2\left(\frac{\pi}{4}\right)}{\left(\frac{\pi}{4}\right)^2} = \frac{1}{2} \cdot \frac{\frac{1}{2}}{\frac{\pi^2}{16}} = \frac{4}{\pi^2}$$

$$F_2 = \frac{A L}{2} \cdot \frac{\sin^2\left(\frac{\pi}{2}\right)}{\left(\frac{\pi}{2}\right)^2} = \frac{1}{2} \cdot \frac{1}{\frac{\pi^2}{4}} = \frac{2}{\pi^2}$$

$$F_3 = \frac{A L}{2} \cdot \frac{\sin^2\left(\frac{3\pi}{4}\right)}{\left(\frac{3\pi}{4}\right)^2} = \frac{1}{2} \cdot \frac{\frac{1}{2}}{\frac{9\pi^2}{16}} = \frac{4}{9\pi^2}$$

$$\begin{aligned} P' &= |F_0|^2 + 2(|F_1|^2 + |F_2|^2 + |F_3|^2) = \\ &= \frac{1}{4} + 2\left(\frac{16}{\pi^4} + \frac{4}{\pi^4} + \frac{16}{81\pi^4}\right) = \\ &= \frac{1}{4} + 2\left(\frac{20}{\pi^4} + \frac{16}{81\pi^4}\right) = \frac{1}{4} + 2 \cdot \frac{1636}{81\pi^4} = \\ &= \frac{1}{4} + \frac{3272}{81\pi^4} \approx 0,6646 \text{ [V}^2\text{]} \end{aligned}$$

(3)

$$\begin{aligned} P' &= \frac{1}{T} \int_0^T x'^2(t) dt = \frac{1}{T} \int_{-\frac{\theta}{2}}^{\frac{\theta}{2}} x'^2(t) dt = \frac{2}{T} \int_0^{\frac{\theta}{2}} x'^2(t) dt = \frac{2}{T} \int_0^{\frac{\theta}{2}} \left(-\frac{2A}{\theta}t + A\right)^2 dt = \\ &= \frac{2}{T} \int_0^{\frac{\theta}{2}} \left(A^2 - \frac{4A^2}{\theta}t + \frac{4A^2}{\theta^2}t^2\right) dt = \frac{2}{T} \cdot \left[A^2 \cdot t \Big|_0^{\frac{\theta}{2}} - \frac{2A^2}{\theta} \cdot t^2 \Big|_0^{\frac{\theta}{2}} + \frac{4A^2}{3\theta^2} \cdot t^3 \Big|_0^{\frac{\theta}{2}} \right] = \\ &= \frac{2}{T} \cdot \left[\frac{A^2 \theta}{2} - \frac{A^2 \theta}{2} + \frac{A^2 \theta}{6} \right] = \frac{A^2 \theta}{3T} = \frac{A^2 L}{3} = \frac{2}{3} = 0,667 \text{ [V}^2\text{]} \end{aligned}$$

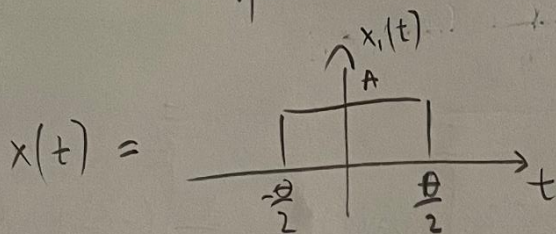
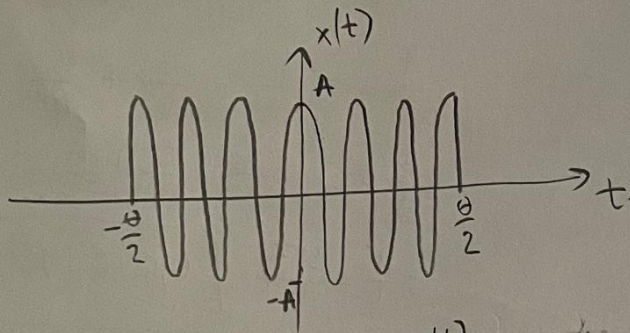
(3)

$$\eta = \frac{P'}{P} = \frac{0,6646}{0,667} \approx 0,996 = 99,6 \%$$

(1)

2.

a)



$$X_{1n} = \frac{A\theta}{T} \cdot \frac{\sin(n\omega_0 \frac{\theta}{2})}{n\omega_0 \frac{\theta}{2}}$$

$$X_1(j\omega) = \lim_{T \rightarrow \infty} T \cdot X_{1n} = A\theta \cdot \frac{\sin(\omega \frac{\theta}{2})}{\omega \frac{\theta}{2}}$$

Hyre: $\omega \frac{\theta}{2} = k\pi$
 $|\omega = \frac{2\pi}{\theta} \cdot k|$

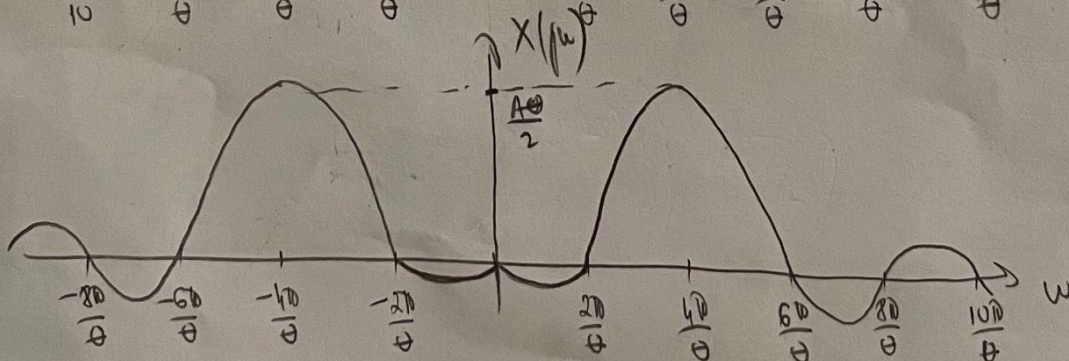
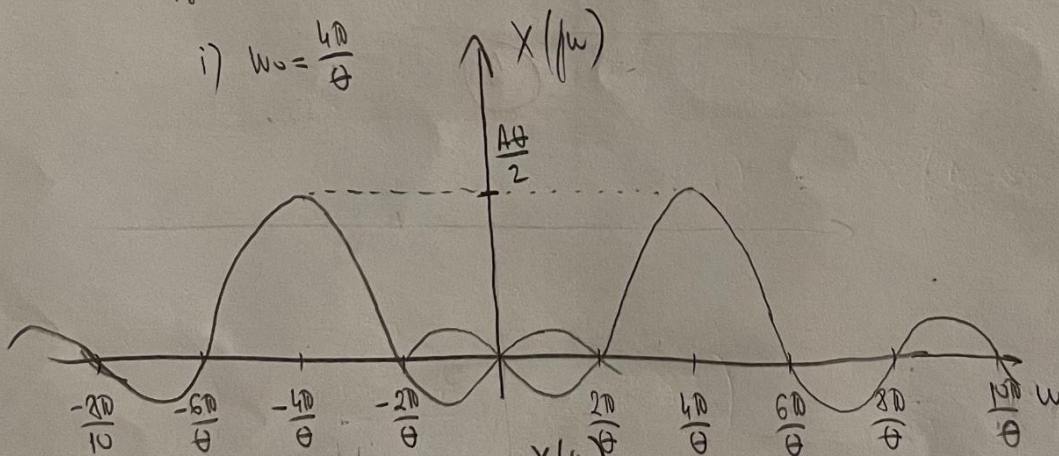
$$x(t) = x_1(t) \cdot \cos(\omega_0 t) = x_1(t) \cdot \frac{1}{2} (e^{j\omega_0 t} + e^{-j\omega_0 t})$$

$$= \frac{1}{2} x_1(t) e^{j\omega_0 t} + \frac{1}{2} x_1(t) e^{-j\omega_0 t}$$

5

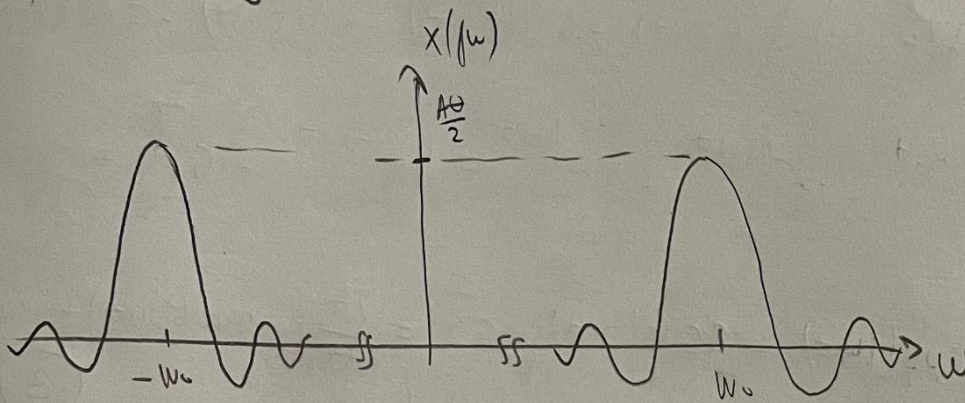
$$X(j\omega) = \frac{1}{2} X_1(\omega - \omega_0) + \frac{1}{2} X_1(\omega + \omega_0)$$

i) $\omega_0 = \frac{4\pi}{\theta}$



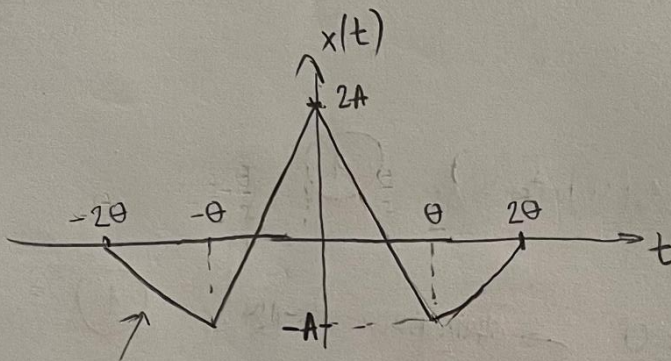
5

ii) $\omega_0 \gg \frac{2\theta}{\theta}$



(5)

8)



1° $(-2\theta, 0)$

$(-\theta, -A)$

$$x_1 = \frac{-A}{\theta} (t + 2\theta) = -\frac{A}{\theta} t - 2A$$

2° $(-\theta, -A)$

$(0, 2A)$

$$x_1 + A = \frac{3A}{\theta} (t + \theta) = \frac{3A}{\theta} t + 2A$$

3° $(0, 2A)$

$(\theta, -A)$

$$x_1 + A = \frac{-3A}{\theta} (t - \theta) = -\frac{3A}{\theta} t + 2A$$

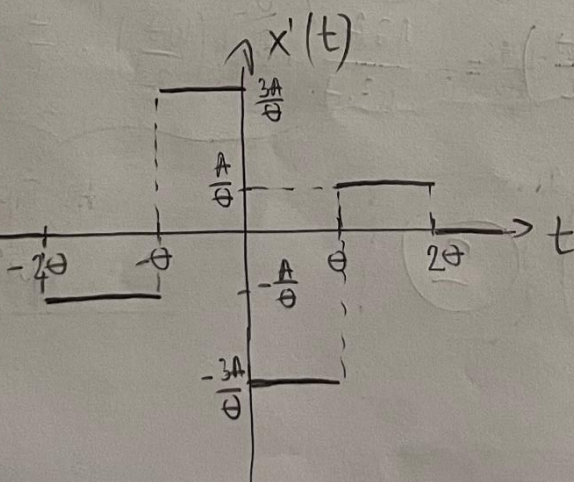
4° $(\theta, -A)$

$(2\theta, 0)$

$$x_1 = \frac{-A}{\theta} (t - 2\theta) = \frac{A}{\theta} t - 2A$$

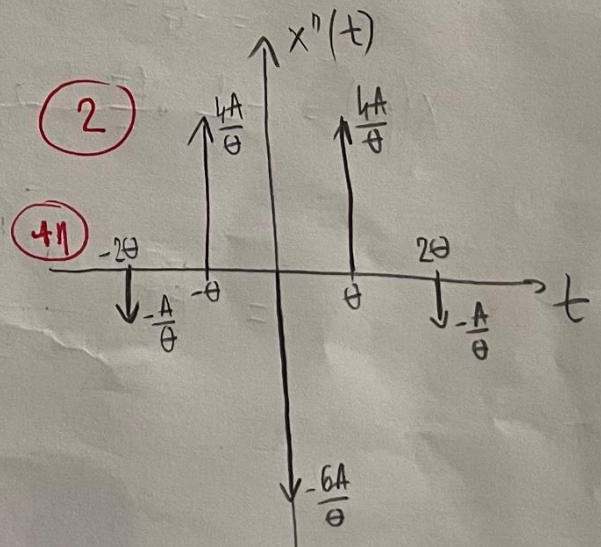
(2)

(+1)



(2)

(+1)



$$\begin{aligned}
 x''(t) = x_2(t) &= -\frac{A}{\theta} \cdot \delta(t+2\theta) + \frac{4A}{\theta} \delta(t+\theta) - \frac{6A}{\theta} \delta(t) + \frac{4A}{\theta} \delta(t-\theta) - \frac{A}{\theta} \delta(t-2\theta) \\
 &= \frac{A}{\theta} [-\delta(t+2\theta) - \delta(t-2\theta) + 4\delta(t+\theta) + 4\delta(t-\theta) - 6\delta(t)] \\
 x_2(j\omega) &= \frac{A}{\theta} [-e^{j\omega 2\theta} - e^{-j\omega 2\theta} + 4e^{j\omega\theta} + 4e^{-j\omega\theta} - 6] = \\
 &= \frac{A}{\theta} [-2\cos(2\omega\theta) + 8\cos(\omega\theta) - 6] = \frac{2A}{\theta} (-\cos(2\omega\theta) + 4\cos(\omega\theta) - 3) \quad (3)
 \end{aligned}$$

$$x_2(0) = 0 \quad (1)$$

$$X_1(j\omega) = \frac{1}{j\omega} \cdot X_2(j\omega) \quad (1)$$

$$X(j\omega) = \frac{1}{j\omega} \cdot X_1(j\omega) + 0 \cdot X_1(0) \cdot \delta(\omega) \quad (1)$$

$$X_1(0) = \int_{-\infty}^{\infty} x_1(t) dt = 0 \rightarrow \text{не парна } \delta - \text{ф-ция!} \quad (1)$$

$$\Rightarrow X(j\omega) = \frac{1}{(j\omega)^2} \cdot X_2(j\omega) = -\frac{1}{\omega^2} \cdot \frac{2A}{\theta} (-\cos(2\omega\theta) + 4\cos(\omega\theta) - 3) = \quad (2)$$

$$\cos 2X = \cos^2 X - \sin^2 X = 2\cos^2 X - 1$$

$$-\cos 2X + 4\cos X - 3 = 1 - 2\cos^2 X + 4\cos X - 3 = -2\cos^2 X + 4\cos X - 2 =$$

$$= -2(\cos^2 X - 2\cos X + 1) = -2 \cdot (1 - \cos X)^2 = -2 \cdot \left(2 \cdot \sin^2 \frac{X}{2}\right)^2 =$$

$$= -8 \cdot \sin^4 \frac{X}{2}$$

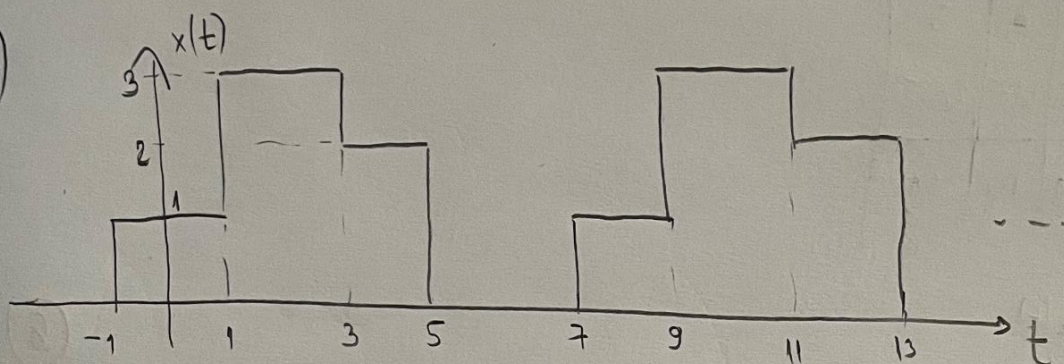
$$X(j\omega) = -\frac{1}{\omega^2} \cdot \frac{2A}{\theta} \cdot \left(-8 \cdot \sin^4 \frac{\omega\theta}{2}\right) = \frac{16A}{\omega^2 \cdot \theta} \cdot \sin^4 \left(\frac{\omega\theta}{2}\right) \cdot \frac{\theta}{4} \cdot \frac{4}{\theta} =$$

$$= 4A\theta \cdot \left(\frac{\sin^2 \left(\frac{\omega\theta}{2}\right)}{\frac{\omega\theta}{2}}\right)^2 \cdot \{2\}$$

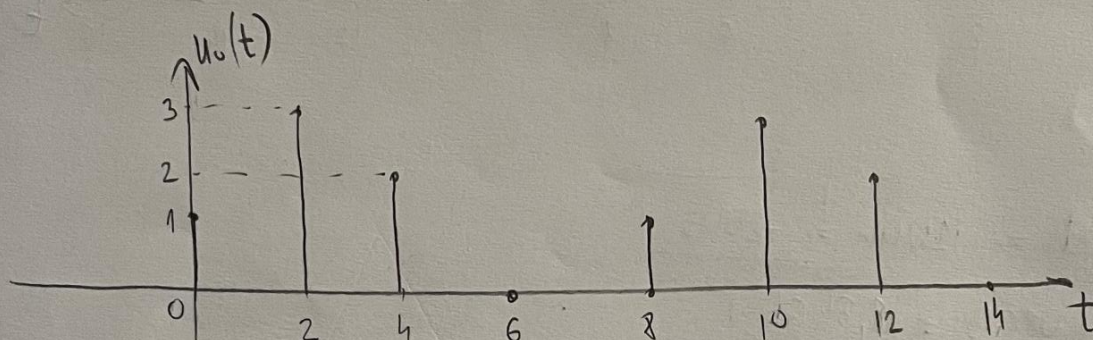
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3.

a)

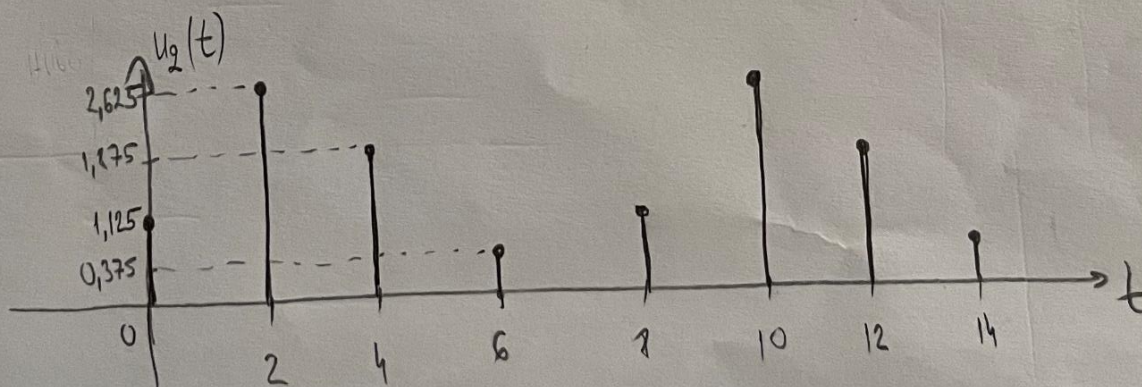
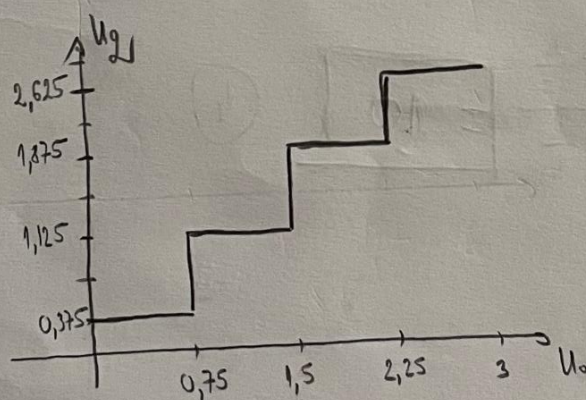


$$T_s = 2s$$

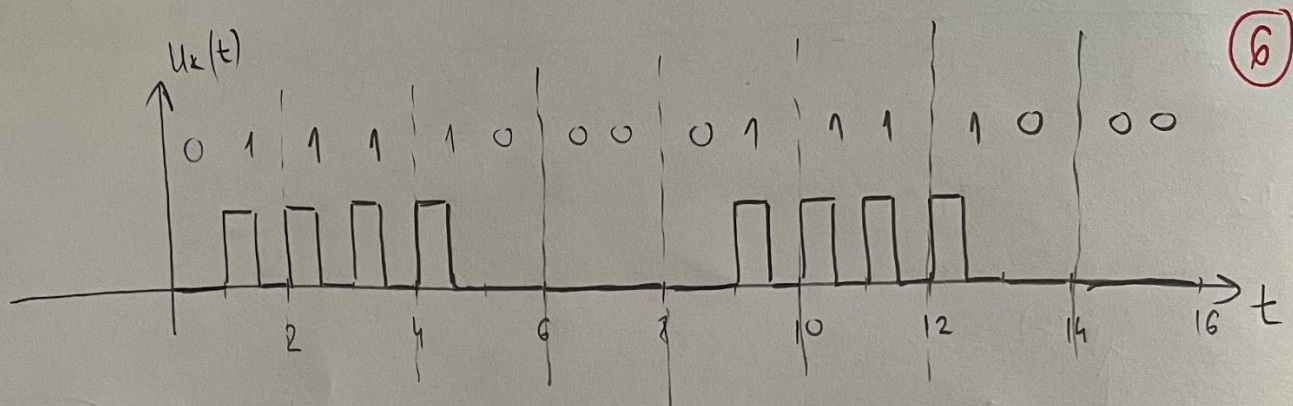


$$\Delta = 0,75V$$

$$\Delta = \frac{x_{\max} - x_{\min}}{2} = \frac{3 - 0}{2} = \frac{3}{2} \Rightarrow 2 = \frac{3}{0,75} \Rightarrow \boxed{4} = 2^4 \Rightarrow \boxed{N=2}$$



Ниво	0	1	2	3
ампл.	0,375	1,125	1,875	2,625
когда пифер	00	01	10	11



b) $t = 2h = 120 \text{ min} = 7200 \text{ s}$

$$J = 1,08 \text{ GB} = 8,64 \text{ Gb}$$

$$v_b' = \frac{J}{t} = \frac{8,64 \cdot 10^9 \text{ b}}{7,2 \cdot 10^3 \text{ s}} = 1,2 \frac{\text{Mb}}{\text{s}} \quad (2)$$

$$v_b' = 1,25 v_b \Rightarrow v_b = \frac{v_b'}{1,25} = 0,96 \frac{\text{Mb}}{\text{s}} \quad (4)$$

$$f_0 = 48 \cdot 10^5 \text{ Hz}$$

$$n = \frac{v_b}{2 \cdot f_0} = \frac{960 \cdot 10^3 \frac{\text{b}}{\text{s}}}{2 \cdot 48 \cdot 10^5 \frac{1}{\text{s}}} = 10 \quad (4)$$